



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Second Semester – 2023

Solved Paper: Probability and Probability Distributions

Roll No.

Course Code: STAT-102

Part-II

Time: 3 Hrs. Marks: 60

Q.1	Briefly answer the following questions
I	Differentiate between Discrete random variable and continuous random variable. Discrete random variable: A discrete random variable is one in which can assume any set of possible values which can be counted or listed. Continuous random variable: A continuous random variable is one which can assume an infinite spectrum of different values across an interval and which cannot be listed.
II	Differentiate between Mutually exclusive events and Collectively exhaustive events. Mutually exclusive events: Two events are said to be mutually exclusive events if they have no common points, i.e $A \cap B = \phi$. For example $A = \{1, 3, 5\}$ and $B = \{2, 4, 6\}$ then $A \cap B = \phi$ Collectively exhaustive events: Events are said to be collectively exhaustive events if they have constitute the entire sample space. For example, $A = \{1, 3\}$, $B = \{2, 4, 6\}$, $C = \{6\}$, $A \cup B \cup C = S$
III	Differentiate between Simple random sampling and Stratified random sampling. Simple random sampling: Simple random sampling is a procedure in which the sample is selected in such a way that every element of a population has a known non-zero (necessarily equal) probability of being included in the sample. Stratified random sampling: Stratified random sampling is a procedure in which the sample is selected from the population which has been divided into a number of non-overlapping groups or sub-populations called strata.
IV	Differentiate between Type-I error and Type-II error: Type-I error: Rejection of null hypothesis when it is true is called type-I error. Type-II error Acceptance of null hypothesis when it is false is called a type-II error.
V	Explain the Central Limit Theorem. Central Limit Theorem: Let $X_1, X_2, X_3, \dots, X_n$ is a random sample of size “n” be drawn from a population having mean = μ and variance = σ^2 . Then the sampling distribution of sample mean ($\bar{X}$) with mean = μ and variance = $\frac{\sigma^2}{n}$ as $n \rightarrow \infty$



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Q.2 Thirty random observations are taken from each of the following distributions and the sample mean is calculated. Find, in each case, the probability that the sample mean exceeds 5.

(a) X is the number of telephone calls made in an evening to a counseling service, where $X \sim P(4.5)$

(b) X is the number of heads obtained when an unbiased coin is tossed nine times.

(a_solution)

It is given that the mean (or variance) of the Poisson distribution = 4.5. As a large sample size, the Poisson distribution tends to Normal distribution. Then

$$P(\bar{X} > 5) = P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{n}} > \frac{5 - 4.5}{\sqrt{\frac{4.5}{30}}}\right)$$

$$P(\bar{X} > 5) = P\left(Z > \frac{5 - 4.5}{\sqrt{0.15}}\right)$$

$$P(\bar{X} > 5) = P(Z > 1.29)$$

$$P(\bar{X} > 5) = 1 - \phi(1.29)$$

$$P(\bar{X} > 5) = 1 - 0.9017$$

$$P(\bar{X} > 5) = 0.098$$

(b_solution)

Mean = $n \cdot p = 9(0.5)$, Variance = $n \cdot p \cdot q = 9(0.5)(0.5) = 2.25$

$$P(\bar{X} > 5) = 1 - P(\bar{X} < 5) = 1 - P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{n}} < \frac{5 - 4.5}{\sqrt{\frac{2.25}{30}}}\right)$$

$$= 1 - P(Z < 1.82) = 1 - \phi(1.82) = 1 - 0.9656 = 0.0344$$



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Q.3 (a) The masses (in grams) of thirteen ball bearings taken at random from a batch had a mean of 429231 and a variance of 5.74. Calculate a 95% confidence interval for the mean of population supposed normal, from which these masses are drawn.

(b) The amount of nicotine, in milligram, in a cigarette of a certain brand is normally distributed with mean μ and standard deviation of 2.5. A random sample of 10 cigarettes yielded a mean nicotine value of 18.4. Test the hypothesis $\mu = 17.8$ against the alternative hypothesis $\mu \neq 17.8$ at the 10% significance level.

(a_solution)

Given that sample mean = 429231, sample S.D = 5.74 and sample size = 13

$$\begin{aligned} \bar{X} \pm t_{\frac{\alpha}{2}(n-1)} \frac{s}{\sqrt{n}} \\ = 429231 \pm t_{0.025(12)} \frac{5.74}{\sqrt{13}} \\ = 429231 \pm (0.51) * (1.5919) \\ = 429231 \pm 0.811869 \\ = (429230.18, 429231.82) \end{aligned}$$

(b_solution)

Procedure for testing

Step.1) State the hypotheses:

$$H_o : \mu = 17.8$$

$$H_o : \mu \neq 17.8$$

Step.2) Level of Significance

$$\alpha = 0.10$$

Step.3) Test statistic:

$$Z = \frac{\bar{X} - \mu}{\sigma}$$

Step.4) Calculation:



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$$Z = \frac{18.4 - 17.8}{\frac{2.5}{\sqrt{10}}} = 0.7589$$

Step.5) Critical region: The tabulated value of Z at 10% level of significance :

$$Z_{\frac{\alpha}{2}} = Z_{\frac{0.10}{2}} = Z_{0.05} = (-1.645, +1.645)$$

Step.6) Our calculated value of Z falls in the acceptance region. So, we fail to reject the null hypothesis.

Q.4 (a) A ball is drawn at random from a box containing 6 red balls, 4 white balls, and 5 blue balls. Determine the probability that the ball drawn is

(i) red ball, $P(\text{red ball}) = \frac{{}^6C_1 * {}^4C_0 * {}^5C_0}{{}^{15}C_1} = \frac{6}{15} = 0.4$

(ii) white ball, $P(\text{White ball}) = \frac{{}^6C_0 * {}^4C_1 * {}^5C_0}{{}^{15}C_1} = \frac{4}{15} = 0.266$

(iii) Blue ball, $P(\text{Blue ball}) = \frac{{}^6C_0 * {}^4C_0 * {}^5C_1}{{}^{15}C_1} = \frac{5}{15} = 0.33$

(iv) Not red ball, $P(\text{not red ball}) = 1 - P(\text{red ball}) = 1 - \frac{{}^6C_1 * {}^4C_0 * {}^5C_0}{{}^{15}C_1} = 1 - \frac{6}{15} = \frac{9}{15}$

(v) Red or White ball, $P(\text{red or white ball}) = P(\text{red ball}) + P(\text{White ball}) = \frac{{}^6C_1 * {}^4C_0 * {}^5C_0}{{}^{15}C_1} +$

$$\frac{{}^6C_0 * {}^4C_1 * {}^5C_0}{{}^{15}C_1}$$

$$P(\text{red or white ball}) = \left(\frac{6}{15} + \frac{4}{15} \right) = \frac{10}{15} = 0.66$$

(b) What is the probability that a card chosen at random from a standard deck of cards will be either a Jack or a Diamond?



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	Total cards: 52 Jack cards: 4 Diamond cards: 13 $P(\text{selected card is a Jack or a Diamond}) = P(J \cap D) = P(J) + P(D) - P(J \cap D)$ $P(\text{selected card is a Jack or a Diamond}) =$ $P(J \cap D) = \frac{{}^4C_1}{{}^{52}C_1} + \frac{{}^{13}C_1}{{}^{52}C_1} - \frac{{}^1C_1}{{}^{52}C_1} = \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52} = 0.30$
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